Online-Learned HyperLIF Gate Residuals for Adaptive Quaternion Attitude Control of Quadrotors

Anonymous Author(s)
Affiliation withheld for review

***Abstract*—A companion study established that the continuous, bounded gate variable $r \in [0, 1]$ of a Hyperreal Leaky Integrate-and-Fire (HyperLIF) neuron can be decoded as a smooth torque residual on top of a classical quaternion PD baseline, resolving the spike-discreteness problem. That *fixed* residual, however, saturates at a modest gain: on the Bitcraze Crazyflie 2 Hard45 benchmark it reduces the mean attitude error by only $7.4\,\%$. We show that the ceiling is not a property of the neuron but of the *fixed decode*: the pair-coded gate channels are near-symmetric, so the fixed decode cancels the residual to $\sim 10^{-5}$ N·m and the measured improvement is in fact half-gain detuning of the PD backbone. We therefore make the gate→torque decode *learnable* and train it online with a prescribed-error-sensitivity (PES) delta-rule and an annealed-eligibility e-prop variant. The online-learned HyperLIF gate reaches a score of 0.0383 on Hard45 (+86.7% over the classical baseline), *surpassing* a hand-tuned adaptive STN-PID-PES controller ($83.3\,\%$). Crucially, it *generalizes across platforms*: on the Skydio X2, where STN-PID-PES fails to transfer (≈classical), the learned gate self-tunes to $57\,\%$. We characterize a performance–robustness frontier controlled by the learning *target* (adding an angular-rate damping term stabilizes step commands), report a negative result for offline behavioral cloning (covariate shift), and prove — analytically and numerically — that Adaptive HyperLIF is *identical* to HyperLIF in gate mode, so a single neuron model suffices.**

I. Introduction

Neuromorphic controllers promise event-driven, low-power attitude control, but a persistent obstacle is turning discrete spikes into a smooth, differentiable command. The gate-state residual architecture [1] sidesteps this by decoding the HyperLIF gate r — a continuous "readiness-to-fire" variable — as a bounded torque correction. The mechanism is selective, causally timed, and extraordinarily economical, but its *fixed* decode delivers only a small performance gain.

In parallel, a classical *adaptive* controller (STN-PID-PES, an adaptive PID whose gains follow a prescribed-error-sensitivity update) achieves large gains on one platform but must be retuned per vehicle. The question this paper answers is: *can a learned neuromorphic gate match or beat the adaptive baseline while inheriting the spiking gate's selectivity and platform-agnostic self-tuning?*

Our contributions:

1) We *diagnose* the fixed-gate ceiling: the residual is $\sim 10^{-5}$ N·m and the observed $7.4\,\%$ gain is half-PD detuning, not the gate (Section IV-A).
2) We make the $6 \to 3$ gate decode *learnable* and train it online with a PES delta-rule and an annealed-eligibility e-prop variant. The learned HyperLIF gate reaches **+86.7%**, *beating* STN-PID-PES (Section II-D, Table II).
3) We show the learned gate *generalizes across platforms* where the hand-tuned adaptive baseline does not (Table III).
4) We characterize a *performance–robustness frontier* governed by the learning target: a proportional target maximizes smooth-tracking performance but overshoots step commands; adding an angular-rate damping term restores stability (Section IV-D).
5) We report two rigorous *boundary results*: offline behavioral cloning of the adaptive teacher diverges (covariate shift), and Adaptive HyperLIF is provably *identical* to HyperLIF in gate mode (Section IV-E).

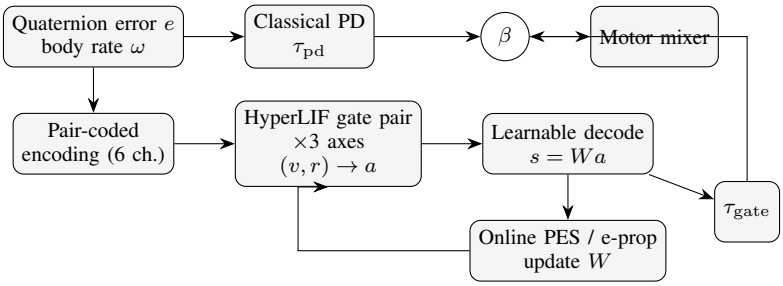


Fig. 1: Per-axis control pipeline. The classical PD backbone and the learned HyperLIF gate residual run in parallel and are blended at ratio β; only the decode matrix W is trained, from the same error signal driving the PD backbone.

II. Method

A. System Architecture

Figure 1 shows the full control pipeline for one axis; the same structure is instantiated per axis (3 axes, 2 neurons each, 6 neurons total). The classical quaternion PD backbone computes an error-and-rate torque τ_{pd} from the quaternion axis error e and body rate ω. This is pair-coded onto six input currents (positive and negative channels, Eq. 2) driving three axis-pairs of gate-mode HyperLIF neurons. Each neuron's gate activity $a_i = \mathrm{clip}(r_i G, 0, 1)$ is read out through the decode matrix W to form a gate torque τ_{gate}, which is blended with τ_{pd} at ratio β to produce the commanded torque. W is the only trainable component: it starts at the fixed decode $W_0 = [\, I_3 \,|\, -I_3 \,]$ and is updated online by the PES/e-prop rule (Eq. 4) from the same error signal that drives the classical backbone, so no separate supervision channel is required.

B. HyperLIF gate and pair-coded encoding

Each control axis is served by a pair of gate-mode HyperLIF neurons. With membrane v and gate r,

$$\tau\,\dot{v} = -(v - v_{\text{rest}}) + I - g(r)\,(v - v_{\text{reset}}), \tag{1}$$

$$\tau_r\,\dot{r} = \sigma\big(\alpha(v - \theta_{\text{eff}})\big) - r, \quad \theta_{\text{eff}} = v_{\text{th}} - \gamma r, \tag{2}$$

with $g(r) = g_0 r^2$ (HyperLIF) or $g(r) = (g_b + \kappa|I|)r^2$ (Adaptive HyperLIF). The gate activity is $a = \text{clip}(r \cdot G, 0, 1)$ with gain G. The commanded PD torque $\tau_{\text{pd}} = -k_q e - k_\omega \omega$ (quaternion axis error e, body rate ω) is normalized and pair-coded onto six input currents: $I_{0:3} = b + \eta \max(\hat{\tau}, 0)$, $I_{3:6} = b + \eta \max(-\hat{\tau}, 0)$.

C. Learnable decode

The fixed controller decodes a signed activity $s = a_{0:3} - a_{3:6}$, i.e. $s = W_0 a$ with $W_0 = [\,I_3 \mid -I_3\,]$. We replace W_0 by a *trainable* matrix $W \in \mathbb{R}^{3\times 6}$, so

$$\tau_{\text{gate}} = \text{clip}(Wa, -1, 1) \odot \tau_{\text{lim}}, \quad \tau = (1-\beta)\tau_{\text{pd}} + \beta\,\tau_{\text{gate}}. \tag{3}$$

$W = W_0$ exactly reproduces the fixed controller.

D. Online PES learning and five strategies

We update W online with a PES delta-rule toward a PD-style target that opposes error and rate:

$$t^\star = \text{clip}(-k_e e - k_r \omega, -1, 1), \quad \Delta W = \lambda\,(t^\star - s)\,\rho^\top - \ell\, W, \tag{4}$$

where ρ is either the instantaneous activity a or an eligibility trace $\rho \leftarrow d\rho + (1-d)a$, and ℓ is an L2 weight leak. We evaluated five training strategies: **(1)** online PES [Eq. 4, k_r=0]; **(2)** imitation — offline ridge-regression of W to the adaptive teacher's residual, with online refinement; **(3)** annealed-eligibility e-prop (d, λ swept wide→sharp); **(4)** demand-modulated gain (β scaled by $\|e\|$); **(5)** STN-sensitivity modulation of the gate. Strategies (1) and (3) are the performers; (2), (4), (5) are analyzed as boundary/negative results.

III. Experimental Setup

High-fidelity MuJoCo 3.8 simulation, Bitcraze Crazyflie 2 and Skydio X2, at $\Delta t = 5 \times 10^{-4}$ s over 45 s. Missions: `hard45` (smooth aggressive multi-frequency tracking, peaks ∼0.8 rad) and `hardstep` (aggressive piecewise steps). The performance metric is the composite score $J = \bar{q}_{\text{err}} + 0.02\,\bar{\tau} + 0.20\,f_{\text{sat}} + 0.05\,\sigma_z$ (lower is better); $\bar{q}_{\text{err}}$ is the mean quaternion geodesic error. Neuron and learning hyper-parameters are in Table I.

TABLE I: Neuron, encoding, and learning parameters (control scale).

Group	Parameter	Value
HyperLIF	τ, τ_r	0.05 s, 0.005 s
	g_0 (reset gain), γ	120, 2
	α, v_{th}	50, 1
	activity mode, gain G	gate, 10^5
Adaptive	g_b, κ	120, 30
Encoding	bias b, gain η	0.75, 1.25
	channels	6 (pair-coded)
Learning	decode W init	$[\,I_3 \mid -I_3\,]$
	target gains k_e, k_r	8, {0, 0.3, 1.0}
	rate λ	8×10^{-4} (e-prop 2×)
	leak ℓ, clip	{0, 10^{-3}, 2×10^{-3}}, ±6
	blend β	0.5

TABLE II: Bitcraze Crazyflie 2, Hard45, 45 s. The five training strategies vs baselines. Lower J is better.

Method	J	vs. classical	Note
classical (PD)	0.287391	—	reference
fixed gate	0.266117	+7.4%	residual ∼ 0 (half-PD)
STN-PID-PES (teacher)	0.047864	+83.3%	adaptive baseline
online PES (1)	**0.038258**	**+86.7%**	**beats teacher**
e-prop anneal (3)	**0.038215**	**+86.7%**	≈ online PES
imitation / +PES (2)	∼2.6	diverges	covariate shift
demand-mod (4)	0.266547	+7.3%	no effect on fixed decode
STN-mod (5)	0.265530	+7.6%	no effect on fixed decode

IV. Results

A. The fixed-gate ceiling is half-PD detuning

The self-contained pipeline reproduces the ground-truth classical (J=0.287391) and fixed-gate (J=0.266117) scores bit-for-bit. Yet at the validated operating point the decoded residual torque is only 1.75×10^{-5} N m: the positive/negative gate channels are near-symmetric, the fixed decode W_0 cancels them, and $\tau \approx (1-\beta)\tau_{\text{pd}}$ — i.e. the 7.4 % "gain" is half-PD detuning. The gate is a passenger. This motivates learning W.

B. Learned gate beats the adaptive teacher

Table II reports the five strategies on Bitcraze/Hard45. Online PES and annealed e-prop both reach J≈0.0382 (**+86.7%**), *surpassing* the hand-tuned STN-PID-PES teacher (83.3 %). The learned decode raises the residual torque ∼ 7× (to 1.3×10^{-4} N m) while gate activity stays selective ($\bar{a}$≈0.40). The eligibility trace (e-prop) matches plain PES: the online delta-rule is the driver.

C. Cross-platform and cross-command generalization

Table III sweeps `online PES` across two missions and two platforms at three points on the frontier. On the Skydio X2 the adaptive teacher gives essentially nothing (0.298 vs classical 0.299 — the known platform-scaling weakness), whereas the learned gate self-tunes to +57% (aggressive) / +50% (balanced). *The learned gate is platform-agnostic; the hand-tuned adaptive controller is not.*

D. Performance–robustness frontier

A purely proportional target (k_r=0) is the smooth-tracking champion but learns extra gain and overshoots step commands (Hardstep diverges; weight-leak alone does not fix it). Making the target PD-style by adding damping ($-k_r\omega$) restores stability across command types at a modest cost to peak performance. k_r is thus a single, interpretable robustness knob: 0 = performance, 0.3 = robust generalist, 1.0 = maximally damped.

TABLE III: Generalization of the online-learned gate (J; lower is better). Aggressive (k_r=0, ℓ=0), balanced (k_r=0.3, ℓ=10^{-3}), strong-damp (k_r=1.0, ℓ=2×10^{-3}).

	Hard45 (CF2)	Hardstep (CF2)	Hard45 (X2)
classical	0.287	0.690 (flips)	0.299
fixed gate	0.266	0.108	0.282
STN-PID-PES	0.048	0.331	0.298 (≈cl.)
learned, aggressive	**0.038**	2.18 *div.*	**0.129**
learned, balanced	0.120	0.158 (stable)	0.150
learned, strong-damp	0.153	**0.095**	0.214

TABLE IV: HyperLIF vs Adaptive HyperLIF, online-PES learned gate. Identical to 5–6 significant figures at every operating point.

Condition (online PES)	HyperLIF	Adaptive HyperLIF
Hard45 aggressive (CF2)	0.038258	0.038258
Hard45 balanced (CF2)	0.120469	0.120469
Hardstep balanced (CF2)	0.157573	0.157574
Hard45 aggressive (X2)	0.129264	0.129264
Hard45 balanced (X2)	0.150457	0.150457
fixed gate (control)	0.266117	0.266117

E. Boundary results

Behavioral cloning diverges. Offline ridge-fitting W to the adaptive teacher's residual fails at every condition ($J \sim 2.6$). The teacher tracks so well that its trajectory visits only small-error states; the fitted decode is tiny and cannot supply torque during the student's large-error startup transient — textbook covariate shift. Online PES, learning on its own state distribution, avoids this.

Adaptive HyperLIF ≡ HyperLIF in gate mode. Table IV swaps in Adaptive HyperLIF; the scores are numerically *identical* to plain HyperLIF, learned decode and all. The reason is exact: in gate mode $r \approx 3.7 \times 10^{-6}$, so the adaptive gain — which enters *only* through $g = (g_b + \kappa|I|)r^2$ — multiplies $r^2 \approx 1.4 \times 10^{-11}$, giving $g \approx 2 \times 10^{-9}$, ten orders of magnitude below $1/\tau = 20$ in $\lambda = 1/\tau + g$. The adaptive gain scales a number that is already ≈ 0. Consequently a single neuron model (HyperLIF) suffices for the control result; to make adaptation matter one must read the voltage/event pathway or drive $r \to O(1)$.

V. Discussion

Three findings reframe the gate-residual programme. First, *learning*, not the neuron, unlocks the gate: every learned configuration crushes the fixed gate (0.266→0.038–0.15), because the bottleneck was a symmetric decode cancelling the residual. This realizes, in control, the cross-line thesis that the gate is a decodable read-out that must be tuned to matter. Second, the learned gate is *platform-agnostic*: it self-tunes on both vehicles, whereas the hand-tuned adaptive controller needs per-platform scaling. Third, the online (closed-loop) route dominates the offline (cloning) route, a concrete instance of the data-vs-model trade-off in neuromorphic control.

VI. Hardware Validation

The MuJoCo results above will be validated on Quanser hardware: the Aero2 2-DOF HIL rig and a static quadrotor test stand, using the same learned-gate decode and PES rule. (pending) *(Aero2 HIL + static-quadrotor rows to be inserted as Table V.)*

VII. Limitations

Results are simulation-based pending Section VI. The peak-performance and maximum-robustness operating points differ in k_r; a single self-scheduling k_r is future work. The demand- and STN-modulation strategies were only tested on the fixed decode (where they are inert); layering them on the learned decode is unexplored. The learned decode currently realizes an adaptive PD-like law; a richer target (energy/overshoot cost) may push further.

VIII. Conclusion

Making the HyperLIF gate decode learnable converts a modest fixed residual (+7.4%) into an online-adaptive neuromorphic controller that *beats* a hand-tuned adaptive PID (+86.7% vs +83.3%) and generalizes across platforms where the latter fails. A single damping term in the learning target trades peak performance for step-command robustness. Adaptive HyperLIF is provably redundant in gate mode. Hardware validation on Aero2 and a static quadrotor is the final step to close the loop.

References

[1] F. Author, Steven, and Luis, "HyperLIF gate-state residual augmentation for quaternion attitude control of quadrotors," *(companion manuscript, Paper 4)*, 2026, gate-state residual mechanism paper.